

Quasi–Monte Carlo Integration on Graphs

Cut Discrepancy, Graph Total Variation, and Spectral Sequences

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Abstract

This note develops coordinate-free analogues of the Koksma–Hlawka inequality for integration over the vertices of a finite weighted graph. The first uses graph total variation and a cut discrepancy, equivalently the minimum congestion of a flow transporting the empirical measure to the target measure. The resulting inequality is an exact norm-duality statement with sharp constant one. A complementary spectral theory pairs graph Sobolev seminorms with negative-Sobolev discrepancies and is better suited to low-frequency and community-level signals. We then give several deterministic equal-weight quadrature constructions: fixed-budget rounded designs, a progressive largest-deficit sequence, a multilevel partition-tree sampler, a spanning-tree specialization with a graph-total-variation guarantee, and a spectral moment-matching sequence for bandlimited graph signals. The relation to classical quasi–Monte Carlo theory and to Hardy–Krause variation is made explicit throughout.

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1 Classical quasi–Monte Carlo theory

Let $P_m = \{x_1, \dots, x_m\} \subset [0, 1]^d$, with repetitions allowed, and let λ denote Lebesgue measure. Classical quasi–Monte Carlo (QMC) approximates

$$I(f) = \int_{[0,1]^d} f(x) d\lambda(x)$$

by the equal-weight rule $Q_m(f) = m^{-1} \sum_{i=1}^m f(x_i)$. Its central deterministic error estimate is the Koksma–Hlawka inequality

$$|Q_m(f) - I(f)| \leq V_{\text{HK}}(f) D_m^*(P_m). \quad (1)$$

Here V_{HK} is Hardy–Krause variation and

$$D_m^*(P_m) = \sup_{t \in [0,1]^d} \left| \frac{1}{m} \sum_{i=1}^m \mathbf{1}_{[0,t)}(x_i) - \lambda([0,t)) \right|$$

is star discrepancy. Thus the error separates into a quantity depending only on the integrand and a quantity depending only on the point set.

In one dimension, V_{HK} is ordinary total variation. In dimensions $d > 1$, however, Hardy–Krause variation contains variation terms on all coordinate faces. This is not an incidental technicality: star discrepancy tests anchored axis-parallel boxes, and the variation is adapted to precisely that family of test sets. Low-discrepancy sequences such as van der Corput, Halton, and Sobol' sequences are designed to balance these boxes at many scales; under suitable constructions their discrepancy decays essentially as m^{-1} up to logarithmic factors [3, 2].

For a general graph there is no canonical family of coordinate boxes. This leaves two principled choices.

- (i) Use a graph-intrinsic family of sets, namely all vertex cuts. This leads to the coordinate-free graph-BV theory in Sections 3–4.
- (ii) Choose a multiscale partition hierarchy, which plays the role of a dyadic coordinate system. This leads to the graph-Sobol construction in Section 5.

2 Graphs, measures, and graph total variation

Let $G = (V, E, w)$ be a finite connected undirected graph, where each edge $e = \{u, v\}$ has a weight $w_e = w_{uv} > 0$. Write $n = |V|$. A probability measure on V is identified with a vector in $\mathbb{R}^V$; hence

$$\int_V f d\mu = \sum_{v \in V} f(v) \mu(v).$$

The target measure μ is arbitrary, subject to $\mu(v) > 0$ where needed below. The uniform choice is $\mu(v) = 1/n$.

Given sampled vertices $v_1, \dots, v_m$, define the empirical measure

$$\nu_m = \frac{1}{m} \sum_{i=1}^m \delta_{v_i}, \quad \eta_m = \nu_m - \mu.$$

Thus $\eta_m(V) = 0$.

Fix an arbitrary orientation of E and let $B \in \mathbb{R}^{E \times V}$ be the oriented incidence matrix. For an oriented edge $e = (u, v)$, $(Bf)_e = f(v) - f(u)$. The graph total variation is

$$\text{TV}_G(f) = \sum_{e \in E} w_e |(Bf)_e| = \sum_{\{u,v\} \in E} w_{uv} |f(u) - f(v)|. \quad (2)$$

The right-hand side is independent of the chosen orientation. It is a seminorm, whose null space consists exactly of the constants because G is connected.

For $\emptyset \neq S \subsetneq V$, define

$$\eta(S) = \sum_{v \in S} \eta(v), \quad w(\partial S) = \sum_{\substack{\{u,v\} \in E \\ u \in S, v \notin S}} w_{uv}.$$

Definition 2.1 (Cut discrepancy). For two probability measures ν and μ on V , define

$$D_{\text{cut},G}(\nu, \mu) = \max_{\emptyset \neq S \subsetneq V} \frac{|(\nu - \mu)(S)|}{w(\partial S)}. \quad (3)$$

The discrepancy in (3) records the worst mass imbalance of a graph region, normalized by the amount of graph connectivity through which that imbalance can be transported. Its normalization is consistent with graph total variation: rescaling all edge weights by a positive constant rescales TV_G and $D_{\text{cut},G}$ inversely, leaving their product unchanged.

3 An intrinsic graph Koksma–Hlawka theorem

We first identify cut discrepancy with a flow norm. This is the basic duality underlying the graph theorem.

Proposition 3.1 (Cut discrepancy is minimum flow congestion). *Let $\eta \in \mathbb{R}^V$ satisfy $\eta(V) = 0$. Then*

$$\min_{B^\top q = \eta} \max_{e \in E} \frac{|q_e|}{w_e} = \max_{\emptyset \neq S \subsetneq V} \frac{|\eta(S)|}{w(\partial S)}. \quad (4)$$

Proof. Write the left-hand side as $\Gamma(\eta)$. If $B^\top q = \eta$, then summing the divergence over S gives

$$\eta(S) = \sum_{e \in \partial S} \sigma_e q_e,$$

where $\sigma_e \in \{-1, 1\}$ depends on the orientation. Therefore

$$|\eta(S)| \leq \max_e \frac{|q_e|}{w_e} w(\partial S).$$

Taking the maximum over S and then the minimum over q proves the inequality $\Gamma(\eta) \geq D_{\text{cut},G}(\eta)$.

For the reverse inequality, fix $t \geq D_{\text{cut},G}(\eta)$ and construct an auxiliary directed network. Replace each undirected edge $\{u, v\}$ by two directed arcs, each of capacity tw_{uv} . Add a source s and an arc $s \rightarrow v$ of capacity $\eta(v)$ whenever $\eta(v) > 0$. Add a sink z and an arc $v \rightarrow z$ of capacity $-\eta(v)$ whenever $\eta(v) < 0$. Let $M = \sum_{\eta(v) > 0} \eta(v)$.

Consider an s – z cut whose source side is $\{s\} \cup S$, for $S \subseteq V$. Its capacity is

$$tw(\partial S) + \sum_{\substack{v \notin S \\ \eta(v) > 0}} \eta(v) + \sum_{\substack{v \in S \\ \eta(v) < 0}} (-\eta(v)) = tw(\partial S) + M - \eta(S).$$

The definition of t implies this is at least M . By the max-flow/min-cut theorem, there is an s – z flow of value M , hence all source arcs are saturated. Cancelling any opposing flows on a pair of directed graph arcs yields a signed flow q on the originally oriented edges. It satisfies $B^\top q = \eta$ and $|q_e| \leq tw_e$. Thus $\Gamma(\eta) \leq t$. Taking the infimum over such t proves (4). $\square$

Theorem 3.2 (Graph total-variation Koksma–Hlawka inequality). *Let G be connected. For every graph signal $f : V \rightarrow \mathbb{R}$ and every pair of probability measures ν and μ on V ,*

$$\left| \int_V f d\nu - \int_V f d\mu \right| \leq \text{TV}_G(f) D_{\text{cut},G}(\nu, \mu). \quad (5)$$

Proof. Set $\eta = \nu - \mu$. By Proposition 3.1, choose a flow q satisfying $B^\top q = \eta$ and

$$\max_e \frac{|q_e|}{w_e} = D_{\text{cut},G}(\nu, \mu).$$

Discrete integration by parts gives

$$\int_V f d\eta = \langle f, B^\top q \rangle = \langle Bf, q \rangle.$$

Hence

$$\left| \int_V f d\eta \right| \leq \sum_{e \in E} |(Bf)_e| |q_e| \leq \left(\sum_{e \in E} w_e |(Bf)_e| \right) \max_{e \in E} \frac{|q_e|}{w_e},$$

which is (5). □

Corollary 3.3 (Exact duality and sharpness). *On the quotient space of graph signals modulo constants, the dual norm of TV_G is $D_{\text{cut},G}$. In particular,*

$$\text{TV}_G(f) = \sup_{\substack{\eta(V)=0 \\ D_{\text{cut},G}(\eta) \leq 1}} \langle f, \eta \rangle.$$

Thus the constant in Theorem 3.2 is sharp.

Proof. Equation (4) identifies $D_{\text{cut},G}$ with the dual of the weighted ℓ^1 norm of Bf . Since B is injective on functions modulo constants, finite-dimensional norm duality gives the claimed formula. □

Remark 3.4 (Disconnected graphs). If G has several connected components, a flow satisfying $B^\top q = \nu - \mu$ exists if and only if $\nu(C) = \mu(C)$ on every component C . Under this condition, Theorem 3.2 applies componentwise. Without it, any variation seminorm that vanishes on componentwise constants cannot control the integration error.

Remark 3.5 (Relation to a Laplacian Sobolev inequality). For a connected graph with Laplacian L , another valid duality estimate is

$$|\langle f, \eta \rangle| \leq \sqrt{f^\top L f} \sqrt{\eta^\top L^\dagger \eta}.$$

It uses the ℓ^2 norm of the graph gradient and an H^{-1} discrepancy. Theorem 3.2 is different: it uses ℓ^1 graph variation and its exact ℓ^∞ flow/cut dual. The former is natural for spectrally smooth signals; the latter is natural for graph-BV signals and mass balance across graph regions.

4 Deterministic equal-weight graph-QMC designs

This section gives elementary constructions that reduce cut discrepancy. Define the weighted inverse vertex-expansion constant

$$\Phi_G = \max_{\emptyset \neq S \subsetneq V} \frac{\min\{|S|, |S^c|\}}{w(\partial S)}. \quad (6)$$

Equivalently, $\Phi_G = 1/h_G$, where h_G is the weighted vertex-expansion constant

$$h_G = \min_{\emptyset \neq S \subsetneq V} \frac{w(\partial S)}{\min\{|S|, |S^c|\}}.$$

Lemma 4.1 (Coordinate imbalance controls cut discrepancy). *Let $N_m(v)$ denote the number of occurrences of v among $v_1, \dots, v_m$, and set*

$$\Delta_m = \max_{v \in V} |N_m(v) - m\mu(v)|.$$

Then

$$D_{\text{cut},G}(\nu_m, \mu) \leq \frac{\Phi_G \Delta_m}{m}. \quad (7)$$

Proof. For $S \subsetneq V$,

$$|\nu_m(S) - \mu(S)| \leq \frac{\Delta_m |S|}{m}.$$

The same bound applied to S^c , together with $(\nu_m - \mu)(S) = -(\nu_m - \mu)(S^c)$, gives

$$|\nu_m(S) - \mu(S)| \leq \frac{\Delta_m \min\{|S|, |S^c|\}}{m}.$$

Divide by $w(\partial S)$ and maximize over S . □

Theorem 4.2 (Fixed-budget rounded design). *For every $m \geq 1$, there exists an m -point multiset such that*

$$D_{\text{cut},G}(\nu_m, \mu) < \frac{\Phi_G}{m}. \quad (8)$$

Consequently, its quadrature error satisfies

$$\left| \int f d\nu_m - \int f d\mu \right| < \frac{\Phi_G}{m} \text{TV}_G(f).$$

Proof. Start with $n_v = \lfloor m\mu(v) \rfloor$ and distribute the remaining $m - \sum_v n_v$ copies to the vertices with largest fractional parts of $m\mu(v)$. Then $\sum_v n_v = m$ and $|n_v - m\mu(v)| < 1$ for every v . Construct the multiset with n_v copies of v . Lemma 4.1 gives (8), and Theorem 3.2 gives the error estimate. □

The design in Theorem 4.2 has the desirable m^{-1} order but is a point set for one prescribed budget. We next give a progressive construction, analogous to a weighted fair-queueing rule.

Theorem 4.3 (Largest-deficit graph-QMC sequence). *Set $N_0(v) = 0$. Given $v_1, \dots, v_m$, choose*

$$v_{m+1} \in \arg \max_{v \in V} ((m+1)\mu(v) - N_m(v)), \quad (9)$$

with any fixed tie-breaking rule. Then, for every $m \geq 1$,

$$\max_{v \in V} |N_m(v) - m\mu(v)| \leq n - 1$$

and hence

$$D_{\text{cut},G}(\nu_m, \mu) \leq \frac{(n-1)\Phi_G}{m}. \quad (10)$$

Proof. Let $r_m(v) = N_m(v) - m\mu(v)$ and suppose inductively that $r_m(v) \leq 1$ for every v . Put

$$a_v = (m+1)\mu(v) - N_m(v) = \mu(v) - r_m(v).$$

Since $\sum_v a_v = 1$, the selected index j in (9) satisfies $a_j \geq 1/n$. Its updated imbalance is

$$r_{m+1}(j) = r_m(j) + 1 - \mu(j) = 1 - a_j \leq 1.$$

For $v \neq j$,

$$r_{m+1}(v) = r_m(v) - \mu(v) \leq r_m(v) \leq 1.$$

The induction begins with $r_0 = 0$, so $r_m(v) \leq 1$ always. Since $\sum_v r_m(v) = 0$, this also implies $r_m(v) \geq -(n-1)$. Thus the asserted coordinate imbalance bound holds, and (10) follows from Lemma 4.1. $\square$

Remark 4.4 (Comparison with Monte Carlo). For independent samples from μ , Hoeffding's inequality and a union bound over all cuts yield a high-probability $m^{-1/2}$ bound, with a graph-size-dependent constant. The deterministic constructions above attain m^{-1} in this finite-graph setting. Their usefulness for a graph family depends on whether the constants Φ_G and, for the progressive rule, $n\Phi_G$, remain controlled in the relevant limit.

5 Partition trees and a graph-Sobol construction

The cut discrepancy in Definition 2.1 is coordinate-free: it tests all graph regions. A second approach deliberately introduces a multiscale coordinate system, just as Sobol' constructions use dyadic boxes.

Let $\mathcal{H}$ be a rooted partition tree. The root is V ; the children of each non-leaf cluster C form a partition of C ; and the leaves are the singletons. For $C \in \mathcal{H}$, write $p(C)$ for its parent and

$$f_C = \frac{1}{\mu(C)} \sum_{v \in C} \mu(v) f(v).$$

Theorem 5.1 (Hierarchical Koksma–Hlawka identity). *For every signed measure η with $\eta(V) = 0$,*

$$\int_V f d\eta = \sum_{\substack{C \in \mathcal{H} \\ C \neq V}} (f_C - f_{p(C)}) \eta(C). \quad (11)$$

Consequently,

$$\left| \int_V f d\eta \right| \leq \left(\sum_{\substack{C \in \mathcal{H} \\ C \neq V}} |f_C - f_{p(C)}| \right) \max_{\substack{C \in \mathcal{H} \\ C \neq V}} |\eta(C)|. \quad (12)$$

Proof. For any leaf v , the increments along the unique root-to-leaf path telescope:

$$f(v) - f_V = \sum_{\substack{C \ni v \\ C \neq V}} (f_C - f_{p(C)}).$$

Because $\eta(V) = 0$, integrating this identity against η gives

$$\int_V f d\eta = \sum_{\substack{C \in \mathcal{H} \\ C \neq V}} (f_C - f_{p(C)}) \sum_{v \in C} \eta(v),$$

which is (11). Taking absolute values proves (12). $\square$

The variation in (12) is a Haar-like multiscale variation adapted to the selected hierarchy. It is not automatically equivalent to TV_G : that equivalence requires additional geometric information about the clusters. Nevertheless, the hierarchy supplies a direct QMC construction.

Proposition 5.2 (Progressive balancing of a partition tree). *At every internal cluster C , route each arriving token to a child C' using the largest-deficit rule with conditional target mass $\mu(C')/\mu(C)$. Let*

$$A_C = \sum_{P \text{ on the path } V \rightsquigarrow p(C)} (\# \text{ch}(P) - 1).$$

Then the resulting leaf sequence satisfies

$$|N_m(C) - m\mu(C)| \leq A_C \quad (C \in \mathcal{H}, m \geq 1). \quad (13)$$

In particular, for a binary partition tree, A_C is at most the depth of C .

Proof. At a node P with $r = \# \text{ch}(P)$ children, the proof of Theorem 4.3, applied to the stream arriving at P , gives

$$\left| N_m(C) - \frac{\mu(C)}{\mu(P)} N_m(P) \right| \leq r - 1$$

for every child C of P . Therefore

$$|N_m(C) - m\mu(C)| \leq r - 1 + \frac{\mu(C)}{\mu(P)} |N_m(P) - m\mu(P)|.$$

Starting from the root, where the discrepancy is zero, induction along the path to C proves (13). $\square$

Combining Proposition 5.2 with Theorem 5.1 yields a direct graph-Sobol type error estimate:

$$\left| \int_V f d\nu_m - \int_V f d\mu \right| \leq \frac{1}{m} \sum_{C \neq V} A_C |f_C - f_{p(C)}|.$$

The natural practical choices for $\mathcal{H}$ are recursive spectral bisections, diffusion-based clusters, or connected multilevel partitions.

5.1 A spanning-tree specialization

The following specialization reconnects the hierarchical construction to intrinsic graph total variation. Let T be a weighted spanning tree of G , with inherited edge weights, and root T at r . Let C_u be the descendant subtree rooted at u .

Lemma 5.3 (Tree cut discrepancy). *For a signed measure η of total mass zero,*

$$D_{\text{cut},T}(\eta) = \max_{u \neq r} \frac{|\eta(C_u)|}{w_{u,p(u)}}.$$

Proof. The right-hand side is attained by the single-edge cut C_u . Conversely, define the unique tree flow by $q_{u,p(u)} = \eta(C_u)$, with sign chosen according to the edge orientation. It has divergence η . Applying the first half of the proof of Proposition 3.1 to this flow shows that every cut ratio is bounded by $\max_{u \neq r} |\eta(C_u)|/w_{u,p(u)}$. $\square$

Corollary 5.4 (Spanning-tree graph-TV QMC). *Generate samples by the partition-tree rule on the rooted subtrees of T , treating a vertex and each descendant subtree as children of the corresponding routing node. Let A_u be the accumulated local balancing constant for C_u . Then*

$$\left| \int_V f d\nu_m - \int_V f d\mu \right| \leq \text{TV}_G(f) \max_{u \neq r} \frac{A_u}{mw_{u,p(u)}}.$$

Proof. Proposition 5.2 gives $|\eta_m(C_u)| \leq A_u/m$. Lemma 5.3 therefore bounds $D_{\text{cut},T}(\eta_m)$. Apply Theorem 3.2 to T , then use $\text{TV}_T(f) \leq \text{TV}_G(f)$ because T is a subgraph of G . $\square$

6 Spectral moment matching

The graph-TV theorem is designed for signals with small edge- ℓ^1 variation. A complementary theory is appropriate when the integrand is well represented by low Laplacian frequencies, as is often the case for low-conductance community-level signals. Let $\mathcal{L}$ be a graph Laplacian self-adjoint in $L^2(\mu)$, with eigenpairs

$$0 = \lambda_0 < \lambda_1 \leq \dots \leq \lambda_{n-1}, \quad \phi_0 = \mathbf{1},$$

and orthonormal eigenvectors ϕ_k . Assume that μ has full support. For a probability measure ν , write

$$\rho_{\nu,\mu}(v) = \frac{\nu(v)}{\mu(v)} - 1, \quad \hat{f}_k = \langle f, \phi_k \rangle_{L^2(\mu)}, \quad \hat{\rho}_k = \langle \rho_{\nu,\mu}, \phi_k \rangle_{L^2(\mu)}.$$

Since $\rho_{\nu,\mu}$ has mean zero, $\hat{\rho}_0 = 0$.

Definition 6.1 (Spectral variation and discrepancy). For $s > 0$, define the homogeneous spectral Sobolev seminorm and its dual discrepancy by

$$\|f\|_{\dot{H}_G^s} = \left(\sum_{k=1}^{n-1} \lambda_k^s |\hat{f}_k|^2 \right)^{1/2}, \quad D_{-s,G}(\nu, \mu) = \left(\sum_{k=1}^{n-1} \lambda_k^{-s} |\hat{\rho}_k|^2 \right)^{1/2}. \quad (14)$$

Theorem 6.2 (Spectral Koksma–Hlawka inequality). For every $s > 0$, every graph signal f , and every probability measure ν on V ,

$$\left| \int_V f d\nu - \int_V f d\mu \right| \leq \|f\|_{\dot{H}_G^s} D_{-s,G}(\nu, \mu). \quad (15)$$

Proof. Because $\rho_{\nu,\mu}$ has mean zero,

$$\int_V f d\nu - \int_V f d\mu = \langle f, \rho_{\nu,\mu} \rangle_{L^2(\mu)} = \sum_{k=1}^{n-1} \hat{f}_k \hat{\rho}_k.$$

Insert $\lambda_k^{s/2} \lambda_k^{-s/2} = 1$ and apply Cauchy–Schwarz:

$$\left| \sum_{k=1}^{n-1} \hat{f}_k \hat{\rho}_k \right| \leq \left(\sum_{k=1}^{n-1} \lambda_k^s |\hat{f}_k|^2 \right)^{1/2} \left(\sum_{k=1}^{n-1} \lambda_k^{-s} |\hat{\rho}_k|^2 \right)^{1/2}.$$

This is (15). □

Remark 6.3 (First-order spectral variation). For $s = 1$, $\|f\|_{\dot{H}_G^1}^2 = \langle f, \mathcal{L}f \rangle_{L^2(\mu)}$. The corresponding discrepancy is the squared $\mathcal{L}^\dagger$ -energy of $\rho_{\nu,\mu}$. This is the ℓ^2 -gradient counterpart of the graph-TV/cut-discrepancy inequality in Theorem 3.2; neither inequality uniformly dominates the other.

Proposition 6.4 (Spectral coefficient Koksma–Hlawka inequality). Let $K < n$ and suppose that $f \in \text{span}\{\phi_0, \dots, \phi_K\}$. For positive numbers $a_1, \dots, a_K$, define

$$V_{K,a}^{(1)}(f) = \sum_{k=1}^K a_k |\hat{f}_k|, \quad D_{K,a}^{(\infty)}(\nu, \mu) = \max_{1 \leq k \leq K} \frac{|\hat{\rho}_k|}{a_k}.$$

Then

$$\left| \int_V f d\nu - \int_V f d\mu \right| \leq V_{K,a}^{(1)}(f) D_{K,a}^{(\infty)}(\nu, \mu). \quad (16)$$

Proof. The constant coefficient makes no contribution to the error. Hence

$$\begin{aligned} \left| \int_V f d\nu - \int_V f d\mu \right| &= \left| \sum_{k=1}^K \widehat{f}_k \widehat{\rho}_k \right| \\ &\leq \sum_{k=1}^K a_k |\widehat{f}_k| \frac{|\widehat{\rho}_k|}{a_k} \\ &\leq V_{K,a}^{(1)}(f) D_{K,a}^{(\infty)}(\nu, \mu). \end{aligned}$$

□

Proposition 6.5 (Community indicators are approximately low frequency). *Suppose that $\mathcal{L} = I - P$ is the random-walk Laplacian of G , where*

$$P(u, v) = \frac{w_{uv}}{d(u)}, \quad \mu(u) = \frac{d(u)}{\text{vol}(V)}, \quad \text{vol}(V) = \sum_{v \in V} d(v).$$

Let $C \subseteq V$, put $f = \mathbf{1}_C$, and let $P_{\leq K}$ be orthogonal projection onto $\text{span}\{\phi_0, \dots, \phi_K\}$. For $0 \leq K \leq n - 2$,

$$\|f - P_{\leq K} f\|_{L^2(\mu)}^2 \leq \frac{w(\partial C)}{\text{vol}(V) \lambda_{K+1}}. \quad (17)$$

Proof. Reversibility gives the Dirichlet-form identity

$$\langle f, \mathcal{L}f \rangle_{L^2(\mu)} = \frac{1}{2} \sum_{u, v \in V} \mu(u) P(u, v) |f(u) - f(v)|^2 = \frac{w(\partial C)}{\text{vol}(V)}.$$

On the other hand,

$$\begin{aligned} \|f - P_{\leq K} f\|_{L^2(\mu)}^2 &= \sum_{k > K} |\widehat{f}_k|^2 \\ &\leq \frac{1}{\lambda_{K+1}} \sum_{k > K} \lambda_k |\widehat{f}_k|^2 \\ &\leq \frac{\langle f, \mathcal{L}f \rangle_{L^2(\mu)}}{\lambda_{K+1}}. \end{aligned}$$

Substitution of the Dirichlet-form identity proves (17). □

Remark 6.6 (Interpretation). For $f = \mathbf{1}_C$, graph total variation is $\text{TV}_G(f) = w(\partial C)$, whereas its first-order spectral energy is $w(\partial C)/\text{vol}(V)$. Thus a community can have a large boundary in absolute weight and still be spectrally simple when that boundary is small relative to graph volume and the relevant spectral scale. Conversely, a large normalized boundary gives no reason to expect a low-frequency representation.

Theorem 6.2 tests all graph frequencies, whereas Proposition 6.4 tests only the first K moments. We now give an equal-weight construction for the latter setting. Let

$$z_K(v) = (\phi_1(v), \dots, \phi_K(v)) \in \mathbb{R}^K, \quad R_K = \max_{v \in V} \|z_K(v)\|_2.$$

Orthogonality gives $\int z_K d\mu = 0$.

Theorem 6.7 (Spectral moment-matching sequence). *For every $K < n$, there is an equal-weight infinite sequence $v_1, v_2, \dots$ supported on at most $K + 1$ vertices such that*

$$\left\| \frac{1}{m} \sum_{i=1}^m z_K(v_i) \right\|_2 \leq \frac{2KR_K}{m} \quad (m \geq 1). \quad (18)$$

Consequently, if $f \in \text{span}\{\mathbf{1}, \phi_1, \dots, \phi_K\}$, then

$$\left| \int_V f d\nu_m - \int_V f d\mu \right| \leq \frac{2KR_K}{m} \left\| f - \int_V f d\mu \right\|_{L^2(\mu)}. \quad (19)$$

Proof. Since $0 = \int z_K d\mu$ belongs to the convex hull of $\{z_K(v) : v \in V\}$, Carathéodory's theorem gives vertices $u_1, \dots, u_r$, where $r \leq K + 1$, and weights $\alpha_j \geq 0$ with

$$\sum_{j=1}^r \alpha_j = 1, \quad \sum_{j=1}^r \alpha_j z_K(u_j) = 0.$$

Apply the largest-deficit rule to the alphabet $\{u_1, \dots, u_r\}$ with target weights (α_j) . If $d_j = N_m(u_j) - m\alpha_j$, the proof of Theorem 4.3 gives $d_j \leq 1$ and $\sum_j d_j = 0$. Hence the sum of positive d_j is at most $r - 1 \leq K$, as is the absolute sum of the negative d_j . Therefore $\sum_j |d_j| \leq 2K$ and

$$\left\| \frac{1}{m} \sum_{i=1}^m z_K(v_i) \right\|_2 = \frac{1}{m} \left\| \sum_{j=1}^r d_j z_K(u_j) \right\|_2 \leq \frac{2KR_K}{m}.$$

This proves (18). If $f = c + \sum_{k=1}^K a_k \phi_k$, its integration error is the inner product of $(a_1, \dots, a_K)$ with the vector on the left-hand side of (18). Cauchy–Schwarz proves (19). $\square$

Remark 6.8 (Approximate bandlimitedness). Let $P_{\leq K}$ be the spectral projector onto $\text{span}\{\phi_0, \dots, \phi_K\}$, let $\rho_m = d\nu_m/d\mu - 1$, and define

$$\|f\|_{\dot{H}_G^s}^2 = \sum_{k \geq 1} \lambda_k^s |\hat{f}_k|^2.$$

The proof above and Cauchy–Schwarz yield

$$\left| \int f d(\nu_m - \mu) \right| \leq \frac{2KR_K}{m} \|P_{\leq K} f\|_{L^2(\mu)} + \|\rho_m\|_{L^2(\mu)} \lambda_{K+1}^{-s/2} \|f\|_{\dot{H}_G^s}.$$

Thus low-mode matching is ideal for bandlimited functions. For general Sobolev functions, K must grow with m and the high-frequency part of the empirical measure must also be controlled. Matching a fixed collection of eigenvectors alone does not control every graph cut.

7 Interpretation and research directions

Theorem 3.2 is the canonical *first-order, coordinate-free* graph analogue of Koksma–Hlawka:

$$\text{integration error} \leq \text{graph total variation} \times \text{cut discrepancy}.$$

It is canonical in the strong sense that the two terms are exact dual norms. Unlike a one-dimensional ordering, it treats graph geometry through flows and cuts. The spectral inequality in Theorem 6.2 is complementary rather than competing: it is adapted to low-frequency signals, while the graph-TV inequality is adapted to edge- ℓ^1 variation and sharp transitions across graph regions.

There is nevertheless no single graph counterpart of every aspect of classical QMC. Hardy–Krause variation is tied to anchored boxes in a chosen product coordinate system. A graph has no intrinsic coordinate faces. A partition tree supplies a deliberate replacement: its clusters are graph analogues of dyadic cells, its tree variation is a graph-Hardy–Krause-type multiscale seminorm, and Proposition 5.2 is a progressive balancing rule analogous to a digital-net construction.

The most promising graph-QMC program is therefore hybrid.

1. Build connected, approximately μ -balanced multiscale graph partitions, for example using Fiedler vectors, diffusion coordinates, or hierarchical clustering.
2. Use progressive quota balancing to make every cluster representative at every scale.
3. Measure both hierarchy discrepancy and intrinsic cut discrepancy.
4. Compare against spectral moment matching, random sampling, and kernel/Leja constructions.
5. Prove graph-family-specific rates in terms of expansion, partition depth, conductance, or spectral growth.

Existing work contains closely related ingredients: graph quadrature based on heat kernels and Laplacian eigenvectors [5], graph Leja sequences based on potential-theoretic kernels [1], and multiscale graph bases built from hierarchical partitions [4]. The cut-discrepancy/graph-TV duality and the progressive equal-weight partition construction isolate a direct route to a graph-QMC theory.

References

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